

PROJECT REPORT

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow

IT Procurement Workflow Automation Project

Project Information	Details
Project Title	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Domain	IT Procurement / Workflow Automation
Automation Platform	Flow-based workflow automation
Project Type	Business Process Automation
Team Members	[Enter team member names and roles]
Guide / Mentor	[Enter guide name]
Institution / Organization	[Enter institution / organization]
Academic Year	2026–2027

Project Report Format

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1. INTRODUCTION

1.1 Project Overview

The project “Streamlining IT Procurement: Automating Standard Laptop Orders with Flow” focuses on automating the repetitive activities involved in requesting and procuring standard laptops. The workflow captures a laptop request, validates required information, checks whether the request matches an approved standard configuration, routes the request through the required approval path, and creates or updates the procurement record automatically.

The objective is to replace manual email-based coordination and spreadsheet tracking with a consistent, traceable and rule-driven process. Flow acts as the orchestration layer connecting the requester, manager, IT/procurement teams and the procurement record.

1.2 Purpose

The purpose of the project is to reduce procurement turnaround time, minimize manual data entry, improve visibility of order status and enforce standard laptop purchasing rules.

- Standardize the laptop request process.
- Automatically validate request details and eligibility.
- Route requests to the correct approver.
- Create procurement tasks/orders without duplicate manual entry.
- Notify stakeholders at important workflow stages.

- Maintain an auditable history of decisions and status changes.

2. IDEATION PHASE

2.1 Problem Statement

Employees who need a standard laptop may currently depend on email, chat messages, spreadsheets or separate procurement forms. This can result in incomplete requests, repeated follow-ups, delayed approvals, inconsistent device selection and limited visibility into the order lifecycle.

The problem addressed by this project is: How can a standard laptop order be processed automatically from request submission to procurement completion while preserving approval controls and status visibility?

2.2 Empathy Map

Perspective	Key Observations
Requester – Says	“I need a laptop quickly and want to know when it will arrive.”
Requester – Thinks	The process should be simple and should not require repeated data entry.
Requester – Does	Submits a request, waits for approval, and follows up for status.
Requester – Feels	Frustrated when the request is delayed or status is unclear.
IT / Procurement	Needs complete data, approved configurations, clear ownership and an audit trail.
Manager / Approver	Needs concise information to make a quick approval decision.

2.3 Brainstorming

- Create a single standard laptop request form.
- Use rules to distinguish standard requests from exceptions.
- Automatically assign approvals based on request attributes.
- Generate procurement tasks or orders automatically after approval.
- Send confirmation and status notifications automatically.
- Escalate overdue approvals or procurement tasks.
- Record timestamps and decisions for reporting and audit.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System / Flow Action	Expected Outcome
Request	Employee submits laptop request.	Validate mandatory fields.	Complete request is created.
Validation	Request waits for processing.	Check standard model, budget and eligibility rules.	Eligible request proceeds automatically.
Approval	Manager reviews request.	Send approval task and track response.	Approved or rejected with reason.
Procurement	Procurement team processes order.	Create/update procurement task or purchase record.	Order enters procurement queue.
Fulfilment	IT prepares device.	Update status and notify requester.	Requester receives fulfilment updates.
Closure	Device is delivered.	Close request and retain audit history.	Completed procurement record.

3.2 Solution Requirements

Functional Requirements

- The system shall provide a standard laptop request form.
- The system shall validate mandatory requester, department, delivery and laptop configuration fields.
- The Flow shall identify whether the requested configuration is part of the approved standard catalogue.
- The Flow shall route requests for approval according to defined business rules.
- Approved requests shall create or update the procurement record automatically.
- The Flow shall notify the requester and relevant teams when the status changes.
- Rejected requests shall record a rejection reason.
- The system shall maintain timestamps and an audit trail for workflow actions.

Non-Functional Requirements

- Reliability: workflow steps should execute consistently without duplicate order creation.
- Security: only authorized users should submit, approve or administer requests.
- Performance: standard requests should progress through automated steps with minimal processing delay.
- Maintainability: business rules and standard catalogue values should be easy to update.
- Usability: requesters should complete the form with minimal training.
- Auditability: all key approvals, status changes and exceptions should be traceable.

3.3 Data Flow Diagram

The following logical data flow can be used as the basis for the implemented Flow:

Step	Flow Component	Data / Action
1	Requester	Submit laptop request
2	Request Form	Capture employee, department, location, model and justification
3	Validation	Check mandatory fields and standard catalogue
4	Approval	Route to manager / designated approver
5	Procurement	Create procurement task or purchase record
6	Notification	Send status updates to requester and stakeholders
7	Fulfilment	IT prepares and assigns the laptop
8	Closure	Mark request complete and store audit history

3.4 Technology / Automation Stack

Layer	Technology / Component	Purpose
User Interface	Request form / portal	Collect standard laptop requirements
Workflow	Flow / workflow engine	Automate rules, approvals, tasks and notifications
Data	Request, employee, catalogue and procurement records	Store workflow information
Business Rules	Conditional logic / decision rules	Validate standard configuration and routing
Notifications	Email / in-platform notifications	Communicate workflow progress
Reporting	Workflow reports / dashboards	Monitor volume, status and turnaround time

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Current Challenge	Proposed Flow Capability
Requests arrive through multiple channels.	Use a single standardized request entry point.

Incomplete information causes rework.	Mandatory fields and validation rules.
Approvals are manually chased.	Automatic approval routing and reminders.
Procurement staff re-enter request details.	Create procurement records from approved request data.
Requester cannot easily see status.	Automatic status notifications and tracking.
Exceptions are difficult to identify.	Conditional branching and exception queues.

4.2 Proposed Solution

The proposed solution is a rule-driven Flow that starts when a standard laptop request is submitted. The Flow validates the request, checks the standard catalogue and approval conditions, routes the request for approval, and creates the procurement work item when approval is granted. Notifications are triggered at submission, approval, procurement and completion stages.

If the request does not meet standard rules—for example, a non-standard model or a value above the defined threshold—the Flow routes it to an exception path instead of creating a normal standard order.

4.3 Solution Architecture

Logical architecture:

Component	Responsibility	Interaction
Requester	Initiates standard laptop request	Submits form to workflow
Request Record	Stores request details and status	Read/write by Flow
Flow Engine	Controls validation, decisions and routing	Reads request and executes actions
Approval Layer	Manager or authorized approver	Approve / reject
Procurement Record	Tracks purchase processing	Created/updated after approval
IT Fulfilment	Prepares and assigns device	Updates fulfilment status
Notification Service	Communicates status	Sends automated messages

High-level flow: Requester → Request Form → Validation → Standard Catalogue Check → Approval → Procurement Record → IT Fulfilment → Closure → Notification.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Deliverable	Suggested Duration
1. Discovery	Understand current procurement process and stakeholders.	Problem definition	1 week

2. Requirements	Define fields, rules, approvals and exceptions.	Requirement specification	1 week
3. Design	Design Flow logic, data mapping and notifications.	Workflow design	1 week
4. Development	Build request form, Flow and procurement actions.	Working automation	2 weeks
5. Testing	Execute functional, negative and performance tests.	Test report	1 week
6. Deployment	User validation, training and production release.	Production workflow	1 week
7. Review	Monitor results and tune rules.	Improvement backlog	Ongoing

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Scenario	Expected Result	Status
TC-01	Submit complete standard laptop request	Request created and Flow starts.	Pass / Update
TC-02	Submit request with missing mandatory field	Submission blocked or validation error shown.	Pass / Update
TC-03	Standard configuration selected	Request follows standard procurement path.	Pass / Update
TC-04	Non-standard configuration selected	Request follows exception path.	Pass / Update
TC-05	Approver approves request	Procurement record is created/updated.	Pass / Update
TC-06	Approver rejects request	Request closes/reverts with rejection reason.	Pass / Update
TC-07	Duplicate trigger condition	No duplicate procurement order is created.	Pass / Update
TC-08	Notification event occurs	Correct stakeholder receives notification.	Pass / Update
TC-09	Procurement status changes	Requester receives updated status.	Pass / Update

TC-10	Workflow completes	Request is closed and audit history retained.	Pass / Update
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Performance measurements should be captured in the actual environment. Recommended measures include workflow execution time, approval turnaround time, procurement processing time, notification delay, error rate and duplicate-order rate. Replace the placeholders with measured values after execution.

7. RESULTS

7.1 Output Screenshots

Insert screenshots from the implemented Flow in this section. Recommended screenshots:

- Standard laptop request form.
- Flow trigger and initial validation step.
- Standard catalogue decision / condition branch.
- Approval action and approval notification.
- Procurement record generated by the Flow.
- Requester status notification.
- Completed request and audit/history view.
- Dashboard or report showing procurement status and turnaround time.

[Insert Screenshot Here]

Figure 1 – Laptop Request Form

[Insert Screenshot Here]

Figure 2 – Flow Logic / Decision Branch

[Insert Screenshot Here]

Figure 3 – Procurement Record

[Insert Screenshot Here]

Figure 4 – Completion / Status View

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Considerations
Reduces repetitive manual work.	Initial workflow design requires analysis and configuration.

Improves request consistency.	Poorly defined rules can route requests incorrectly.
Speeds up standard approvals and procurement.	Workflow failures require monitoring and support.
Provides better status visibility.	Changes to procurement policy may require rule updates.
Creates an audit trail.	Integration or notification dependencies can affect automation.
Reduces duplicate data entry.	Exception cases still need controlled manual handling.

9. CONCLUSION

The project demonstrates how a Flow-based automation can streamline standard laptop procurement by connecting request intake, validation, approvals, procurement processing, notifications and closure into one controlled workflow. The approach improves process consistency and visibility while reducing repetitive administrative effort.

The solution is designed so that standard requests can move through a predictable path, while non-standard requests are redirected to an exception process. This separation supports automation without removing necessary procurement controls.

10. FUTURE SCOPE

- Integrate directly with vendor catalogues and purchase-order systems.
- Add automatic stock and inventory availability checks before ordering.
- Introduce SLA timers, reminders and escalation for delayed approvals.
- Use analytics to measure procurement cycle time and identify bottlenecks.
- Add automated asset registration after laptop delivery.
- Support employee onboarding workflows that trigger laptop provisioning.
- Introduce policy-based approval thresholds that can be maintained without redesigning the entire Flow.
- Add exception analytics to identify frequently requested non-standard configurations.

11. APPENDIX

A. Standard Laptop Request Fields

Field	Example / Description	Required
Requester	Employee submitting the request	Yes
Employee ID	Unique employee identifier	Yes
Department	Requester's department	Yes
Location	Delivery / office location	Yes

Laptop Model	Approved standard model	Yes
Configuration	RAM, storage, processor, etc.	Yes
Business Justification	Reason for laptop request	Yes
Manager	Approving manager	Yes
Required By Date	Expected delivery date	Yes
Exception Reason	Required only for non-standard request	Conditional

B. Sample API / Integration Documentation

If the selected Flow platform exposes APIs or integrates with external procurement systems, document the actual endpoints used by the implementation. The following is a documentation template rather than a claim about a specific platform.

Operation	Purpose	Method / Interface	Authentication
Create Request	Create laptop procurement request	[Enter actual method/interface]	[Enter method]
Get Catalogue	Retrieve approved standard laptop models	[Enter actual method/interface]	[Enter method]
Create Procurement Record	Create purchase/procurement task	[Enter actual method/interface]	[Enter method]
Update Status	Update procurement / fulfilment status	[Enter actual method/interface]	[Enter method]
Send Notification	Notify requester or approver	[Enter actual method/interface]	[Enter method]

C. Authentication and Authorization

Access to the request, approval and procurement functions should be restricted according to organizational roles. Requesters should be able to submit and view their own requests; approvers should be able to act on assigned approvals; procurement and IT users should have access to the records required for fulfilment; administrators should manage workflow configuration. Document the actual authentication mechanism, role model, tokens or sessions used by the implemented platform.

D. Setup / Deployment Instructions

1. Create or configure the required request, catalogue, approval and procurement data structures.
2. Configure the standard laptop catalogue and business rules.
3. Build the request form and map each field to the request record.
4. Configure the Flow trigger and validation actions.

5. Configure standard versus exception decision branches.
6. Configure approval routing and notifications.
7. Configure procurement record creation or integration.
8. Run the test cases and record actual results.
9. Deploy the approved Flow to the production environment.

E. Source Code / Configuration (if any)

[Insert repository link, Flow export/package, configuration file reference, or state “Not applicable” if the solution is configured entirely through the Flow platform.]

F. Dataset / Reference Links

Dataset / catalogue reference: [Insert link or internal reference]

GitHub / Project Demo Link: [Insert link if applicable]